Roadmap. Srednicki §1: why relativity needs fields. pp. 1–8: the square-root and Klein–Gordon equations fail as one-particle wave equations; the metric and Lorentz transformations serve all course. Dirac returns in LecNote16–18. pp. 8–13: demote $\mathbf{x}$ to a label; n-body QM as a field theory. p. 14 (unassigned) begins $[V, N]$, zero by p. 12's corrected rule. Next: LecNote2 (§2) and LecNote3 (§3). Supplements: QFTHistory, LorentzGroup, FieldTransformations.

Section #1. Class 1

Attempts in Relativistic QM.

Axioms: < Schrödinger Equation.

States are represented by vectors in Hilbert Space

Observables by Hermitian operators

Measurements yield one of the eigenvalues

Identical particles, etc.

$$i\hbar \frac{\partial}{\partial t} |\psi, t\rangle = \mathcal{H} |\psi, t\rangle .$$

let's consider the simplest

$$\mathcal{H} = \frac{1}{2m} \vec{p}^2$$

$$i\hbar \frac{\partial}{\partial t} \psi(\vec{x}, t) = -\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{x}, t)$$

$|\psi, t\rangle$ in Griffith was the general vector in Hilbert Space $|S(t)\rangle$. and $\psi(x,t) = \langle x | S(t) \rangle$ $= \int d^3y\, \delta(x-y)\psi(y,t)$ wave function.

$\psi(x,t) = \langle x | \psi, t \rangle$ is the position space

$= \psi(x,t)$ We want to generalize into relativistic version

$$\mathcal{H} \sim E \longrightarrow \mathcal{H} = \sqrt{p^2c^2 + m^2c^4}$$

$$\mathcal{H} = mc^2 + \frac{1}{2m}p^2 + \cdots$$

↑ constant ; $\frac{1}{2m}p^2$ ↑ QM. ; ↑ corrections.

Now SE becomes.

$$-i\hbar\frac{\partial}{\partial t}\psi(x,t) = +\sqrt{m^2c^4 + \hbar^2c^2\nabla^2}\,\psi(x,t)$$

Still treats space-time differently.
non-linear so doing it is problematic[1] in general.

[1] Linear in ψ; "non-linear" means in p (Dirac's is "linear" on p. 8). The trouble is non-locality: $\sqrt{m^2c^4 - \hbar^2c^2\nabla^2}$ acts through a kernel, $-m^2K_2(mr)/2\pi^2r^2$ at $\hbar = c = 1$, nonzero at every separation r and decaying over $\hbar/mc$. So $\partial_t\psi(\mathbf{x})$ depends on ψ everywhere, and t (first order) and $\mathbf{x}$ (all orders) are treated unequally.

How about we "Square" it?

$$-\hbar^2\frac{\partial^2}{\partial t^2}\psi(x,t) = (m^2c^4 + \hbar^2c^2\nabla^2)\psi(x,t)$$

~~If $x = (x,t)$.~~ $x^\mu = (ct, \vec{x})$.

and a contravariant[1]

$$x_\mu = (-ct, \vec{x}) \qquad \mu = 0, 1, 2, 3.$$

Minkowski metric.

$$g_{\mu\nu} = \begin{pmatrix} -1 & & & \\ & +1 & & \\ & & +1 & \\ & & & +1 \end{pmatrix}.$$

$$x_\mu = g_{\mu\nu} x^\nu.$$

the inverse.

$$x^\mu = g^{\mu\nu} x_\nu.$$

requires $g^{\mu\nu} = \begin{pmatrix} -1 & & & \\ & +1 & & \\ & & +1 & \\ & & & +1 \end{pmatrix}$.

so we have

$$g^{\mu\nu} g_{\nu\rho} = \delta^\mu_{\ \rho}. \quad \text{the Kronecker Delta.}$$

$$= \begin{matrix} \mu \\ \\ \\ \\ \end{matrix} \overset{\rho}{\begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}}$$

1 Should read *covariant*: lower-index $x_\mu = g_{\mu\nu} x^\nu = (-ct, \mathbf{x})$ is the covariant vector; upper-index $x^\mu = (ct, \mathbf{x})$ (end of p. 2) is the contravariant one.

Special Relativity tells us that physics looks the same in all inertial frames.

x^μ specify a inertial frame.

another coordinates $\bar{x}^\mu$ can be related to it by

$$\bar{x}^\mu = \underbrace{\Lambda^\mu{}_\nu}_{\text{const}} x^\nu + \underbrace{a^\mu}_{\text{const.}}$$

↓ Lorentz transformation

↘ translation vector.

Furthermore, we need to require that.

$$g_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma = g_{\rho\sigma}$$ (Recall proper time).

which keeps the length unchanged.

"interval" between two points

as

$$(x - x')^2 = g_{\mu\nu} (x - x')^\mu (x - x')^\nu$$
$$= (x - x')^2 - c^2 (t - t')^2$$

$$(\bar{x}-\bar{x}')^2 = g_{\mu\nu}(\bar{x}-\bar{x}')^\mu(\bar{x}-\bar{x}')^\nu$$

$$= g_{\mu\nu}\Lambda^\mu{}_\rho\,\Lambda^\nu{}_\sigma\,(x-x')^\rho(x-x')^\sigma$$

$$= g_{\rho\sigma}(x-x')^\rho(x-x')^\sigma$$

$$= (x-x')^2 \quad \text{as desired.}$$

Now Special relativity Equivalents demands

$$\mathcal{L}(x) = \mathcal{L}(\bar{x})$$

and obey same equations in their own frame.

Lets define

$$\partial_\mu = \frac{\partial}{\partial x^\mu} = \left(\frac{1}{c}\frac{\partial}{\partial t}, \vec{\nabla}\right).$$

$$\partial^\mu = \frac{\partial}{\partial x_\mu} = \left(-\frac{1}{c}\frac{\partial}{\partial t}, \vec{\nabla}\right).$$

$$\partial^\mu x^\nu = g^{\mu\nu}.$$

Since $\bar{x}^\mu = \Lambda^\mu{}_\nu x^\nu + a^\mu$.

$$\bar{\partial}^\mu = \Lambda^\mu{}_\nu \partial^\nu.$$

we can easily check that.

$$\bar{\partial}^\rho \bar{x}^\sigma = g^{\rho\sigma}$$

as

$$\bar{\partial}^\rho \bar{x}^\sigma = \Lambda^\rho{}_\nu \partial^\nu (\Lambda^\sigma{}_\mu x^\mu + a^\sigma)$$

$$= \Lambda^\rho{}_\nu \partial^\nu x^\mu \Lambda^\sigma{}_\mu$$

$$= \Lambda^\rho{}_\nu \Lambda^\sigma{}_\mu g^{\mu\nu} = g^{\rho\sigma}.$$

Now $\partial^2 \equiv \partial^\mu \partial_\mu = -\frac{1}{c^2}\frac{\partial}{\partial t^2} + \vec{\nabla}^2.$

K.G. $(-\partial^2 + m^2c^2/\hbar^2)\psi(x) = 0.$

And in $\bar{x}$ inertial frame

$$(-\bar{\partial}^2 + m^2c^2/\hbar^2)\psi(\bar{x}) = 0$$ [1]

Now

1 $\bar{\psi}$ is the field in barred coordinates: same value at the same event, $\bar{\psi}(\bar{x}) = \psi(x)$ (p. 5). The barred KG is for $\bar{\psi}(\bar{x})$; with $\bar{\partial}^2 = \partial^2$ (p. 7) it is the same equation. Above, $\Lambda^\rho{}_\nu \Lambda^\sigma{}_\mu g^{\nu\mu} = g^{\rho\sigma}$ follows from p. 4: $\Lambda^T g \Lambda = g$ gives $\Lambda^{-1} = g^{-1}\Lambda^T g$, so $\Lambda g^{-1}\Lambda^T = g^{-1}$ (Srednicki (2.6)).

we need to show that the two K.G

in two frames are the same,

$$\bar{\partial}^2 = g_{\mu\nu}\bar{\partial}^\mu\bar{\partial}^\nu = g_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma\partial^\rho\partial^\sigma$$

$$= g_{\rho\sigma}\partial^\rho\partial^\sigma = \partial^2$$

so these are two identical equations.

But K.G Equation doesn't obey SE.

and the equation changes from first order to second order.

then the normalization of states

$$\langle\psi,t|\psi,t\rangle = \int d^3x \langle\psi,t|\vec{x}\rangle\langle\vec{x}|\psi,t\rangle$$

$$= \int d^3x\, \psi^*(x)\psi(x)$$

is not time independent.

thus probability is not conserved

one can see more details by working out $\frac{d}{dt}\langle\ \rangle$ (K.G doesn't help here since it is 2nd order derivative)

1 Should read $\bar{\partial}^2 = g_{\mu\nu}\bar{\partial}^\mu\bar{\partial}^\nu$, both indices up, as the next step uses $\bar{\partial}^\nu = \Lambda^\nu{}_\sigma\partial^\sigma$ (p. 6). As written, $g_{\mu\nu}\bar{\partial}^\mu\bar{\partial}_\nu = \sum_\nu \bar{\partial}_\nu\bar{\partial}_\nu$ is not invariant.

2 Second order: ψ and $\partial_t\psi$ at $t=0$ are independent, so $\frac{d}{dt}\int|\psi|^2 = 2\,\mathrm{Re}\int\psi^*\partial_t\psi$ is not fixed. For $\psi = e^{i\mathbf{k}\cdot\mathbf{x}}(Ae^{-i\omega t} + Be^{i\omega t})$, $|\psi|^2$ oscillates at 2ω. KG conserves $\rho = i(\psi^*\partial_t\psi - \partial_t\psi^*\psi) = 2\omega(|A|^2 - |B|^2)$ instead ($\hbar = c = 1$), negative for negative frequencies: not a probability (LorentzGroup).

$$H_{ab} = cP_j(\alpha^j)_{ab} + mc^2(\beta)_{ab}$$

Skipping Dirac's effort. as we'll be introducing it with a fresh eye of symmetries & Representations later this course.

linear & conserves probability.

found anti particle states and Dirac sea.

negative Energy.

And its already about infinite # of particles.

but still we can't describe bosons.

we need a pause. and now we realise that

Time is not yet an Hermitian operator. (skipped Jan. 22. 25)

so we never treated it properly.

Two operations { denote $\vec{x}$ to a label.

promote t to an operator.

When promote $t \longrightarrow T$ operator.

one can use the proper time τ in SE.

coordinate time, in Heisenberg Picture.

$\hat{X}^\mu(\tau)$. has $\hat{X}^0 = T$.

many complications, any monotonic function of τ

is good here so is finite redundancy.

Also one can consider more parameters

$\hat{X}^\mu(\sigma, \tau)$ now classically this gives

a family of world lines that we call world sheets.

$\hat{X}^\mu(\sigma,\tau)$ is a propagating String

beyond ~~our~~ our scope here.

Another option is to denote $x_\cdot$, as labels on operators.

consider an operator $\varphi(\vec{x})$.

A set of such operators are call quantum fields.

$\varphi(\vec{x}) \longrightarrow$

Heisenberg Picture

$\varphi(\vec{x}, t) = e^{iHt/\hbar}\, \varphi(\vec{x}, 0)\, e^{-iHt/\hbar}$.

so now $\vec{x}, t$ are labels on operators $\varphi(\vec{x}, t)$.

none is an eigenvalue of an operator.

↑ Quantum Field.

These two options are equivalent.

but $\varphi(\vec{x}, t)$ corresponds to Quantum Field Theory.

is much more convenient for most problems which

is the subject of this course.

For example of classical field theory, see, e.g. Tong, Schwartz, etc.

There is another desired property,

Normal fixed # QM can be written as

non-~~relativistic~~

QFT.

Let's begin with a position space S.E. for n particles.

$$i\hbar\frac{\partial}{\partial t}\psi = \left[\sum_{j=1}^{n}\left(-\frac{\hbar^2}{2m}\vec{\nabla}_j^2 + U(\vec{x}_j)\right) + \sum_{j=1}^{n}\sum_{k=1}^{j-1} V(x_j - x_k)\right]\psi$$

Here $\psi = \psi(\vec{x}_1, \vec{x}_2, \vec{x}_3, \ldots \vec{x}_n; t)$

the QM can be rewritten by introducing a quantum field $a(\vec{x})$ and $a^+(\vec{x})$ to have

$$[a(\vec{x}), a(\vec{x}')] = 0$$

$$[a^+(\vec{x}), a^+(\vec{x}')] = 0$$

$$[a(\vec{x}), a^+(\vec{x}')] = \delta^3(\vec{x} - \vec{x}')$$

$a^+(x)$ and $a(x)$ is like SHO's creation & annihilation operators.

We can introduce a new $\mathcal{H}$ for our QFT.

$$\mathcal{H}_{QFT} = \int d^3x\, a^+(\vec{x})\left(-\frac{\hbar^2}{2m}\vec{\nabla}^2 + U(\vec{x})\right)a(\vec{x}) + \frac{1}{2}\int d^3x\, d^3y\, V(\vec{x}-\vec{y})\, a^+(\vec{x})\, a^+(\vec{y})\, a(\vec{y})\, a(\vec{x})$$

Now consider a state

$$|\psi, t\rangle = \int d^3x_1\, d^3x_2 \cdots d^3x_n\, \psi(\vec{x}_1, \vec{x}_2, \ldots \vec{x}_n; t)\, a^+(\vec{x}_1)\, a^+(\vec{x}_2) \cdots a^+(\vec{x}_n)|0\rangle$$

here $|0\rangle$ is the vacuum state that

$$a(x)|0\rangle = 0.$$

We can show that. SE of $\mathcal{H}$ QFT is satisfied if and not[1] if

$\mathcal{H}$ $\psi(\vec{x}_1, \vec{x}_2, \ldots \vec{x}_n; t)$

satisfy SE of $\mathcal{H}$.

[1] If and only if: $a(\mathbf{x})a^\dagger(\mathbf{x}_1)\cdots a^\dagger(\mathbf{x}_n)|0\rangle = \sum_j \delta^3(\mathbf{x}-\mathbf{x}_j)\prod_{k\neq j} a^\dagger(\mathbf{x}_k)|0\rangle$, so the $a^\dagger(\ldots)a$ term puts $-\frac{\hbar^2}{2m}\nabla_j^2 + U(\mathbf{x}_j)$ on each $\mathbf{x}_j$, and $\frac{1}{2}\iint V a^\dagger a^\dagger a a$ gives $\sum_{j<k} V$. Only the symmetric part of ψ survives: bosons. Fermions need anticommutators (LecNote3 p. 1).

so they are equivalent

and. $a^\dagger(\vec{x}_1)|0\rangle$ create a particle at position $\vec{x}_1$

can be interpreted as

the operator $N = \int d^3x\, a^\dagger(\vec{x}) a(\vec{x})$ counts the total # of particles.

$[N, \mathcal{H}] = 0$ so particle # is conserved.[2]

↑ e.g.
$[a(x), a^\dagger(y)a(y)]$
$= \delta^3(x-y)$
$[a^\dagger(x), a^\dagger(y)a(y)]$
$= -\delta^3(x-y)$
so even powers cancel & commutes
And $\mathcal{H}$ acts on $\mathcal{H}$

[2] Should read $[a(x), a^\dagger(y)a(y)] = \delta^3(x-y)\,a(y)$ and $[a^\dagger(x), a^\dagger(y)a(y)] = -\delta^3(x-y)\,a^\dagger(y)$, so $[a(x), N] = a(x)$, $[a^\dagger(x), N] = -a^\dagger(x)$. Terms with as many $a^\dagger$ as a commute with N; not "even powers": $[N, a^{\dagger 3}a] = 2a^{\dagger 3}a$.

but for a general theory, e.g.

$\mathcal{H} \supset \int d^3x\, [a^\dagger(\vec{x}) a(\vec{x}) + \text{h.c.}]$

QM will fail[1] and QFT prevail.

There are also requirements for bosons & fermions, different that we will study in near future classes.

1 Why QM fails: by p. 12's rule, $[N, a^{\dagger 2}a] = a^{\dagger 2}a$ and $[N, a^\dagger a^2] = -a^\dagger a^2$, so this term changes N by ± 1 and $[N, H] \neq 0$. One particle evolves into a superposition of 1, 2, 3, ... particles: no $\psi(\mathbf{x}_1 \dots \mathbf{x}_n; t)$ with fixed n describes it, but a state in Fock space does. Relativity makes such terms unavoidable (QFTHistory).

$$[V, V] = \int d^3x d^3y \; V(\vec{x}-\vec{y})\, a^\dagger(x) a^\dagger(y) a(y) a(x) \cdot \int d^3z \, a^\dagger(z) a(z)$$

$$\int d^3x d^3y d^3z \Big[V(\vec{x}-\vec{y})\, a^\dagger(x) a^\dagger(y) a(y) a(z) \cdot a^\dagger(z) a(x) - a^\dagger(z) a(z) V(\vec{x}-\vec{y}) a^\dagger(x) a^\dagger(y) a(y) a(x) \Big]$$